RFI DETECTION IN SENTINEL-1 SAR IMAGERY VIA PSEUDO-LABEL DISTILLATION AND MULTI-RESOLUTION ENSEMBLE

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ABSTRACT

We describe the *UdeM AI* team's solution to the ESA ClearSAR Track 1 Grand Challenge: Radio-Frequency Interference (RFI) detection in Sentinel-1 SAR quicklooks. The pipeline combines a YOLO26x base detector with a pixel-space Normalised Wasserstein Distance (NWD) regression loss, pseudo-label distillation from a multi-source ensemble teacher onto the unlabelled test imagery, multi-resolution test-time augmentation (TTA) fused by Weighted Box Fusion (WBF), and a selective cross-fold ensemble that drops the domain-overlapping fold. The final submission reaches 0.4776 $\text{mAP}_{50\text{-}95}$ on the held-out test set, 28th of 172 teams. We report two measurements. On a leak-isolated meta-validation set the distilled student source gains $+0.077$ $\text{mAP}_{50\text{-}95}$ over a cold baseline; on the held-out test set the same source is redundant with the ensemble teacher, and the full distillation–TTA–blend stack adds only $+0.0021$ over the teacher alone. The gap between the two is the main result.

***Index Terms*—** Ensemble, pseudo-label distillation, radio-frequency interference, SAR imagery, small object detection.

1. TASK AND DATA

The ESA ClearSAR Track 1 Grand Challenge [1] frames RFI detection in Sentinel-1 quicklooks as a COCO-style object detection task with mAP averaged over IoU 0.50–0.95. The labelled set contains 3,154 images; 786 test images are held out by the organisers and scored by the challenge platform from a zipped COCO-format JSON of detections. RFI appears as thin horizontal stripes (median height $\approx$10 px, median width $\approx$140 px) in variable-size RGB quicklooks (median $\approx 515\times342$ px), with a median per-box aspect ratio of $\approx$8:1. Three constraints follow: IoU-based regression gives no gradient for non-overlapping thin boxes; SAR side-looking geometry breaks horizontal symmetry, so flip TTA is disabled (H-flip costs -6.5% [2]); and inference resolutions above 640 px amplify speckle relative to signal. The five-fold split is used to *construct* diverse ensemble members, not as an out-of-fold generalisation estimator: cross-fold validation mAPs are leaked (each fold detector has seen the other folds' held-out images) and are never used for model selection. We evaluate on two separate sets: a 401-image leak-isolated meta-validation set sampled from the fold-0 validation pool (seed 42), used as a single-model gate, and the official held-out test set, used for model selection and final reporting.

2. METHOD

The pipeline (Figure 1) has four stages.

Base detector. YOLO26x [3] ($\approx$59 M parameters, COCO pre-trained, no P2 head) trained with a CIoU + NWD box loss $\mathcal{L}_{\text{box}} = (1-\alpha)\mathcal{L}_{\text{CIoU}}+\alpha\mathcal{L}_{\text{NWD}}$ at α=0.5, with NWD [4] computed in pixel space (normalisation constant C=12.8 px). Ultralytics evaluates the box loss on coordinates normalised to the network input, so we rescale the positive boxes to pixel units before computing NWD; otherwise C is far too large, the similarity saturates, and the gradient vanishes in the small-height regime characteristic of RFI ($\approx$10 px median). Images are pre-processed with CLAHE (clip 3.0, 8×8 tiles) on the LAB L-channel. Horizontal and vertical flips are disabled. Training uses MuSGD for up to 60 epochs (patience 15) at 640 px, batch 32, η_0=10^{-3} decaying linearly to 10^{-5}.

Pseudo-label distillation. The teacher (V12) is our strongest pre-distillation ensemble: a single-stage WBF [5] over multiple YOLO26x fold and multi-resolution-TTA sources together with one RF-DETR-Large [6] fold checkpoint at weight $1.5\times$, fused with τ=0.70 and `conf_type`=max. V12 scores 0.4755 on the held-out test set. The teacher is applied to the 786 unlabelled test images with confidence threshold 0.75 (max 20 boxes/img), producing 600 pseudo-labelled images. For each fold $f\in\{1,2,4\}$, a student is fine-tuned from the per-fold base on the fold-augmented dataset with AdamW [7] (η_0=10^{-4}, `freeze`=10, 15 epochs, batch 24, same NWD blend). Fold 0 is omitted because its base partition overlaps the meta-validation pool.

Multi-resolution TTA and fusion. Each student is inferred at $\mathcal{I}=\{608,640,672,704\}$ px; WBF across resolutions per fold yields one TTA source. The three per-fold TTA sources are then fused with equal weights into a super-source S^*. The final submission V17 is the two-source WBF blend $D_{\text{final}} = \text{WBF}(D_{\text{V12}}, S^*; w{=}(1.0, 0.25), \tau{=}0.70)$.

3. RESULTS AND DISCUSSION

Table 1(a) decomposes the student source: pseudo-distillation is the largest single component ($+0.040$), as the soft signal in the teacher's confidence scores compensates for the small training set; multi-resolution TTA adds $+0.035$ because thin stripes a few pixels tall are resolved unevenly by YOLO's strided feature pyramid at any single input size; dropping the domain-overlapping fold-0 is net-positive because correlated predictions drag the WBF consensus. Small-box accuracy tracks the same chain (AP_S $0.356\rightarrow0.418$); AP_{75} stays the weakest column (0.522 vs 0.752 at AP_{50}), placing the residual error at high IoU. Table 1(b) measures transfer: on the held-out test set the full stack adds only $+0.0021$ over the teacher and $+0.0056$ over a plain 5-fold ensemble. The $+0.077$ student source is redundant with the teacher, which already encodes the same multi-fold knowledge; the meta-validation ceiling (0.515) is itself unreachable there because V12 is leaked on that split. Pushing further regresses: blend weight w=0.375 gives 0.4768, and a second-generation pseudo-FT blend gives 0.4566, below the teacher. Intermediate stages were scored only

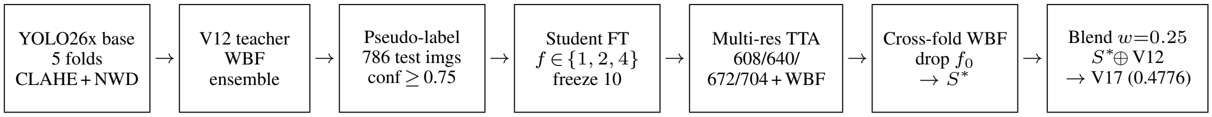

Fig. 1. Pipeline. A YOLO26x base detector (CLAHE input, pixel-space NWD loss) is trained per fold; the V12 pre-distillation ensemble pseudo-labels the 786 test images; per-fold students are fine-tuned on the fold-augmented data, run multi-resolution TTA fused by WBF, fused across folds (dropping the domain-overlapping fold 0) into a super-source S^*, and blended with V12 at w=0.25.

Table 1. **(a)** Construction of the student super-source S^* on the 401-image leak-isolated meta-validation set (single-source; reproducible from archived predictions). **(b)** Contribution on the official held-out test set, the only leak-free measurement for the full pipeline (V12 is leaked on meta-validation).

(a) Meta-validation (student source only)		
Configuration	**mAP$_{50\text{-}95}$**	Δ
Cold fold-0 baseline	0.394	—
Pseudo-FT student, fold 4	0.434	+0.040
+ Multi-res TTA (4 res, WBF)	0.469	+0.035
+ 3-fold WBF (drop f_0) = S^*	0.471	+0.002
(b) Held-out test (full pipeline)		
Configuration	**mAP$_{50\text{-}95}$**	Δ
Plain 5-fold WBF ensemble	0.4720	—
Teacher V12	0.4755	+0.0035
Full pipeline V17	**0.4776**	+0.0021

on meta-validation, as official submissions were rate-limited to one per 12 h.

Negative results. A second architectural family (RF-DETR) plateaued at $+0.0002$ across six checkpoint–weight variants; D-FINE-X [8] fine-tuned on fold 0 for 10 epochs reached 3×10^{-4} mAP, suggesting DETR-family decoders with different matching [9] would need substantially more epochs to adapt to the elongated geometry of RFI; inference-only SAHI [10] at 320-px tiles regressed -0.085, fragmenting long horizontal stripes at tile boundaries; and temperature-scaled calibration was at most $+0.0002$.

4. CONCLUSION

The final pipeline reaches 0.4776 mAP$_{50\text{-}95}$ on the held-out test set, placing the *UdeM AI* team 28th of 172 teams in ESA ClearSAR Track 1, on a single cloud NVIDIA A100 GPU. On unseen SAR data a quality-weighted multi-fold ensemble teacher recovers most of the attainable accuracy, and the distillation–TTA–blend stack adds little beyond it. The largest remaining gap is at high IoU thresholds, where sub-pixel height accuracy dominates and a 640-px input under-resolves the median ≈10-px stripe; sliced *training* at 320-px tiles is the most promising next step.

5. REFERENCES

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